

Soln: **Claim:** $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False if and only if the common domain consists of at most 3 elements.

Proof: **Direction 1:** Common domain consists of at most 3 elements $\Rightarrow \forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False.

Case 1: Common domain: $\{a\}$
 $(a \neq a)$ is False. By defn. of AND, $(a \neq a) \wedge (a \neq a) \wedge (a \neq a)$ is False.
By defn. of existential quantifier, $\exists w ((w \neq a) \wedge (w \neq a) \wedge (w \neq a))$ is False.
By defn. of universal quantifier, $\forall z \exists w ((w \neq a) \wedge (w \neq a) \wedge (w \neq z))$ is False.
By defn. of universal quantifier, $\forall y \forall z \exists w ((w \neq a) \wedge (w \neq y) \wedge (w \neq z))$ is False.
By defn. of universal quantifier, $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False.

Case 2: Common domain: $\{a_1, a_2\}$
 $(a_1 \neq a_1)$ is False. By defn. of AND, $(a_1 \neq a_1) \wedge (a_1 \neq a_1) \wedge (a_1 \neq a_2)$ is False.
 $(a_2 \neq a_2)$ is False. By defn. of AND, $(a_2 \neq a_2) \wedge (a_2 \neq a_1) \wedge (a_2 \neq a_2)$ is False.
By defn. of OR, $((a_1 \neq a_1) \wedge (a_1 \neq a_1) \wedge (a_1 \neq a_2)) \vee ((a_2 \neq a_1) \wedge (a_2 \neq a_1) \wedge (a_2 \neq a_2))$ is False.
By defn. of existential quantifier, $\exists w ((w \neq a_1) \wedge (w \neq a_1) \wedge (w \neq a_2))$ is False.
By defn. of universal quantifier, $\forall z \exists w ((w \neq a_1) \wedge (w \neq a_1) \wedge (w \neq z))$ is False.
By defn. of universal quantifier, $\forall y \forall z \exists w ((w \neq a_1) \wedge (w \neq y) \wedge (w \neq z))$ is False.
By defn. of universal quantifier, $\forall x \forall y \forall z \exists w ((w \neq x) \wedge (w \neq y) \wedge (w \neq z))$ is False.

Case 3: Common domain: $\{a_1, a_2, a_3\}$
 $(a_1 \neq a_1)$ is False. By defn. of AND, $(a_1 \neq a_1) \wedge (a_1 \neq a_2) \wedge (a_1 \neq a_3)$ is False.
 $(a_2 \neq a_2)$ is False. By defn. of AND, $(a_2 \neq a_1) \wedge (a_2 \neq a_2) \wedge (a_2 \neq a_3)$ is False.
 $(a_3 \neq a_3)$ is False. By defn. of AND, $(a_3 \neq a_1) \wedge (a_3 \neq a_2) \wedge (a_3 \neq a_3)$ is False.
By defn. of OR, $((a_1 \neq a_1) \wedge (a_1 \neq a_2) \wedge (a_1 \neq a_3)) \vee ((a_2 \neq a_1) \wedge (a_2 \neq a_2) \wedge (a_2 \neq a_3)) \vee ((a_3 \neq a_1) \wedge (a_3 \neq a_2) \wedge (a_3 \neq a_3))$ is False.